Week 7 - Graded Assignment 7

The due date for submitting this assignment has passed.
Due on 2026-04-01, 23:59 IST.
You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-01, 07:55 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1 point

Consider the vector spaces VV and functions $\langle .,. \rangle : V \times V \rightarrow R$⟨.,.⟩:V×V→R defined as follows:

i) $V = R^2$V = R2 and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 - x_2 y_1 - x_1 y_2 + 2x_2 y_2$.⟨(x1, x2), (y1, y2)⟩ = x1y1 − x2y1 − x1y2 + 2x2y2.

ii) $V = M_{2\times 2}(R)$V = M2×2(R) and $\langle A, B \rangle = Tr(AB)$, ⟨A, B⟩ = Tr(AB),
where $Tr(M)$Tr(M) denotes the trace of a matrix MM, i.e., the sum of the diagonal elements of MM.

iii) $V = M_{2\times 1}(R)$V = M2×1(R) and $\langle A, B \rangle = Tr(AB^t)$, ⟨A, B⟩ = Tr(ABt),
where $Tr(X)$Tr(X) denotes the trace of a matrix XX, i.e., the sum of the diagonal elements of XX. Y^tY t denotes the transpose of matrix YY.

iv) $V = R^2$V = R2 and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 + x_2 y_1$.⟨(x1, x2), (y1, y2)⟩ = x1y2 + x2y1.

- [x] (i) is an inner product.
- [x] (ii) is an inner product.
- [ ] (iii) is an inner product.
- [ ] (iv) is an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(i) is an inner product.
(iii) is an inner product.

1 point

Consider two linear transformations T and ST and S from $R^2 \rightarrow R^2$R2 → R2 defined as $T(x, y) = (2x + y, x + y)$ and $S(x, y) = (x + cy, x + 2y)$T(x, y) = (2x + y, x + y) and S(x, y) = (x + cy, x + 2y). Let A and BA and B be matrix representations of linear transformations T and S T and S with respect to the standard bases of R^2R2 respectively.
Consider the following statements:

- **P:** If c = 1c = 1, then A and BA and B are similar matrices.
- **Q:** If c = 2c = 2, then A and BA and B are similar matrices.
- **R:** If c = 1 and $P^{-1}AP = B$c = 1 and P−1AP = B, then PP can be the matrix $\begin{bmatrix}10\\12\end{bmatrix}.\begin{bmatrix}1102\end{bmatrix}$.
- **S:** If c = 1 and $P^{-1}AP = B$c = 1 and P−1AP = B, then PP can be the matrix $\begin{bmatrix}12\\11\end{bmatrix}\begin{bmatrix}1121\end{bmatrix}$.
- **T:** If c = 1c = 1, then there are infinitely many PP satisfying the equation $P^{-1}AP = B$P−1AP = B.

Which of the following options are true?

- [x] PP is true but QQ is false.
- [ ]

Both P and QP and Q are true.

Both R and SR and S are true.

RR is false but SS is true.

TT is true.

Yes, the answer is correct.

Score: 1

Accepted Answers:

PP is true but QQ is false.

RR is false but SS is true.

TT is true.

1 point

Let $\langle .,. \rangle \langle .,. \rangle$⟨.,.⟩⟨.,.⟩ denote the standard inner product on R^2 R2, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$ ⟨(x1, x2), (y1, y2)⟩ = x1y1 + x2y2. Which one of the following options is (are) true for the vector $\gamma \in R^2$ γ ∈ R2, such that $\langle \alpha, \gamma \rangle = 4$⟨α, γ⟩ = 4 and $\langle \beta, \gamma \rangle = 8$⟨β, γ⟩ = 8, where $\alpha = (3, 1)$α = (3, 1) and $\beta = (6, 2)$β = (6, 2).

No such γγ exists.

There are infinitely many such vectors which satisfy the properties of γγ.

γγ is unique in R^2 R2.

Any vector in the set $\{(t, 4 - 3t) \mid t \in R\}$ {(t, 4 − 3t) ∣ t ∈ R} satisfies the properties of γγ.

$(1, 1)$(1, 1) is the only vector which satisfies the properties of γγ.

Yes, the answer is correct.

Score: 1

Accepted Answers:

There are infinitely many such vectors which satisfy the properties of γγ.

Any vector in the set $\{(t, 4 - 3t) \mid t \in R\}$ {(t, 4 − 3t) ∣ t ∈ R} satisfies the properties of γγ.

1 point

Let $\langle .,. \rangle \langle .,. \rangle$⟨.,.⟩⟨.,.⟩ denote the standard inner product on R^2 R2 , i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$ ⟨(x1, x2), (y1, y2)⟩ = x1y1 + x2y2 and let $\nu \in R^2$ ν ∈ R2. Consider a linear transformation $T_\nu : R^2 \rightarrow R$Tν : R2 → R defined as:

$$T_\nu(u) \text{Tν(u)} = \langle u, \nu \rangle \langle \text{u, ν} \rangle,$$

where $\nu \in R^2$ ν ∈ R2. Which of the following options is (are) true for T_ν Tν?

T_ν Tν is one-one for all $\nu \neq 0 \in R^2$ ν ≠ 0 ∈ R2 $= 0 \in$ R2.

T_ν Tν is onto for all $\nu \neq 0 \in R^2$ ν ≠ 0 ∈ R2 $= 0 \in$ R2.

T_ν Tν is onto for all $\nu \in R^2$ ν ∈ R2.

T_ν Tν is not one-one for every $\nu \in R^2$ ν ∈ R2.

There exists a $v \in R^2$ v ∈ R2 such that T_v Tv is not onto.

Yes, the answer is correct.

Score: 1

Accepted Answers:

T_v Tv is onto for all $v \neq 0 \in R^2$ v = 0 ∈ R2.

T_v Tv is not one-one for every $v \in R^2$ v ∈ R2.

There exists a $v \in R^2$ v ∈ R2 such that T_v Tv is not onto.

1 point

Let A and B be square matrices of the same order nn. Which of the following statements are sufficient to conclude that A is equivalent to B.

- [ ] det(A) = det(B).
- [x] Nullity(A) = Nullity(B).
- [ ] The set of columns of AA, and the set of columns of BB are both linearly dependent subsets of R^n Rn.
- [x] There exists a matrix CC such that AA is equivalent to CC, and BB is equivalent to CC.
- [x] There exist invertible matrices PP and QQ of order nn such that $PA = BQ$PA = BQ.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Nullity(A) = Nullity(B).

There exists a matrix CC such that AA is equivalent to CC, and BB is equivalent to CC.

There exist invertible matrices PP and QQ of order nn such that $PA = BQ$PA = BQ.

1 point

Let LL and L' L′ be affine subspaces of R^3 R3, where $L = (0, 1, 1) + U$L = (0, 1, 1) + U and $L' = (0, 1, 0) + U'$ L′ = (0, 1, 0) + U′, for some vector subspaces UU and U' U′ of R^3 R3. Let a basis for UU be given by $\{(1, 1, 0), (1, 0, 1)\}$ {(1, 1, 0), (1, 0, 1)} and a basis for U' U′ be given by $\{(1, 0, 0)\}$ {(1, 0, 0)}. Suppose there is a linear transformation $T: U \to U'$ T : U → U′ such that $(1, 0, 1) \in \ker(T)$ (1, 0, 1) ∈ ker(T) and $T(1, 1, 0) = (1, 0, 0)$T(1, 1, 0) = (1, 0, 0). An affine mapping $f: L \to L'$ f : L → L′ is obtained by defining $f((0, 1, 1) + u) = (0, 1, 0) + T(u)$f((0, 1, 1) + u) = (0, 1, 0) + T(u), for all $u \in U$u ∈ U. Which of the following options are true?

- [x] $L = \{(x, y + 1, x - y + 1) \mid x, y \in R\}$L = {(x, y + 1, x − y + 1) ∣ x, y ∈ R}.
- [ ] $L' = \{(x, y + 1, 0) \mid x, y \in R\}$L′ = {(x, y + 1, 0) ∣ x, y ∈ R}.
- [ ] $L = \{(x - y, y + 1, x - y + 1) \mid x, y \in R\}$L = {(x − y, y + 1, x − y + 1) ∣ x, y ∈ R}.
- [ ] $L = \{(x, x + 1, y + 1) \mid x, y \in R\}$L = {(x, x + 1, y + 1) ∣ x, y ∈ R}.
- [x] $f(x, y + 1, x - y + 1) = (y, 1, 0)$f(x, y + 1, x − y + 1) = (y, 1, 0)
- [ ] $f(x - y, y + 1, x - y + 1) = (x, y + 1, 0)$f(x − y, y + 1, x − y + 1) = (x, y + 1, 0)
- [ ] $f(x, x + 1, y + 1) = (y, 1, 0)$f(x, x + 1, y + 1) = (y, 1, 0)
- [ ] $f(x, y + 1, x - y + 1) = (0, 1, y)$f(x, y + 1, x − y + 1) = (0, 1, y)

Yes, the answer is correct.

Score: 1

Accepted Answers:
$L = \{(x, y + 1, x - y + 1) \mid x, y \in R\}$L = {(x, y + 1, x − y + 1) ∣ x, y ∈ R}.
$f(x, y + 1, x - y + 1) = (y, 1, 0)$f(x, y + 1, x − y + 1) = (y, 1, 0)
Let θθ be the angle between the vectors $u = (4, 7, 3)$u = (4, 7, 3) and $v = (1, 2, -6)$v = (1, 2, −6), then what will be the value of $cos(\theta)$cos(θ)?

0

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 0
1 point
Consider a basis

$$\{v_1 = (1, 2, 0), v_2 = (2, -1, 0), v_3 = (0, 0, 2)\}\{v1 = (1, 2, 0), v2 = (2, -1, 0), v3 = (0, 0, 2)\}$$

of R^3 R3 with usual inner product. Suppose $v = (x, y, \frac{3x + y}{5}) \in V$v = (x, y, 53x+y) ∈ V is written as $v = c_1 v_1 + c_2 v_2 + c_3 v_3$v = c1v1 + c2v2 + c3v3, such that $c_1 + c_2 = 4$c1 + c2 = 4. What will be the value of c_3 c3?

2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 2
1 point

Let $V = R^2$ V = R2 be a vector space. Consider two inner products $\langle ., . \rangle_1$ ⟨., .⟩1 and $\langle ., . \rangle_2$ ⟨., .⟩2 on VV defined as

$$\langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1 x_2 - x_1 y_2 - x_2 y_1 + 4 y_1 y_2 \langle (x1, y1), (x2, y2) \rangle 1 = x1x2 - x1y2 - x2y1 + 4y1y2$$

and

$$\langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3x_1 x_2 + 2 y_1 y_2 . \langle (x1, y1), (x2, y2) \rangle 2 = 3x1x2 + 2y1y2.$$

If $\langle (a, b), (4, 3) \rangle_1 = 35 \langle (a, b), (4, 3) \rangle 1 = 35$ and $\langle (a, b), (4, 3) \rangle_2 = 60 \langle (a, b), (4, 3) \rangle 2 = 60$, then find the value $a + 2b$a + 2b.

11

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 11
1 point

Let $V = R^3$ V = R3 be the inner product space with usual inner product. If θθ is the angle between $(15, 8, 18)$(15, 8, 18) and (a, b, c)(a, b, c) where $15a + 8b + 18c = 0$15a + 8b + 18c = 0 and $(a^2 + b^2 + c^2) \neq 0$(a2 + b2 + c2) = 0, then find the value of $\cos\theta$cos θ.

0

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 0

1 point

Consider a vector v = $(x - 20, 2, 1) \in R^3$ v = (x − 20, 2, 1) ∈ R3. Find the value of xx so that the length of the vector vv is minimum.

20

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) 20
1 point

Consider the following subsets of R^3 R3, with $\langle , \rangle$⟨ , ⟩ used to denote the standard dot product on R^3 R3, and $\|\|$‖‖ used to denote the standard length function.
S_1 = $\{v \in R^3 \mid \langle v, (0, 0, 1)\rangle = 2, \|v\| = 1\}$$S_2$ = $\{u \in R^3 \mid u \in \text{Span}(\{(1, 1, 1)\}), \|u\| = 1\}$S1 = {v ∈ R3 ∣ ⟨v, (0, 0, 1)⟩ = 2, ‖v‖ = 1}S2 = {u ∈ R3 ∣ u ∈ Span({(1, 1, 1)}), ‖u‖ = 1}
If n_1 n1 and n_2 n2 denote the number of elements in S_1 S1 and S_2 S2, respectively, then find the value of $n_1 - n_2$ n1 − n2.

-2

Yes, the answer is correct.
Score: 1
Accepted Answers:
(Type: Numeric) -2
1 point